Quantum Algorithms for Biomedical and Translational Applications

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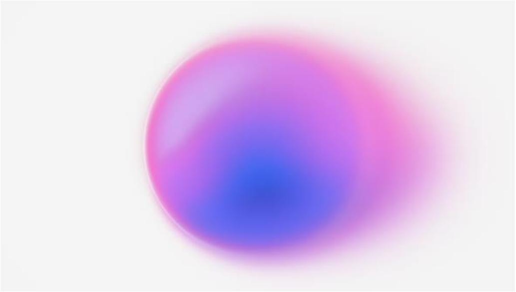
What is Quantum Computing?

Computer that exploits quantum mechanical properties to process information



- Superposition
- A quantum system existing in multiple states simultaneously until it is measured

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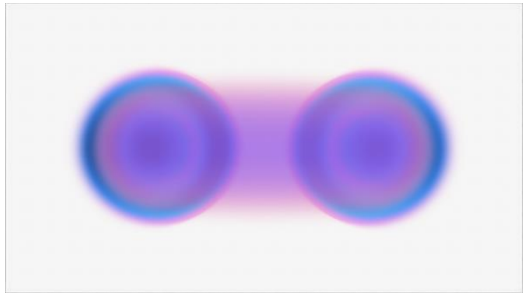
Parallelism



- Entanglement
- Information shared jointly between entangled pairs or groups

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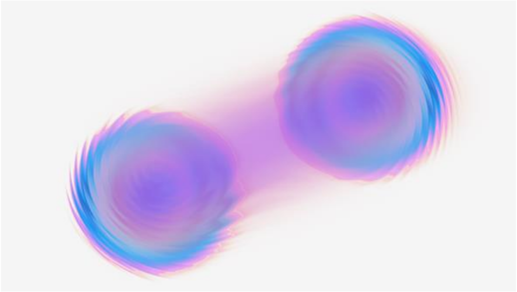
Correlation



- Interference
- Interaction that affects likelihood of solutions

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Convergence

IBM Quantum: System Two (Utility Scale)

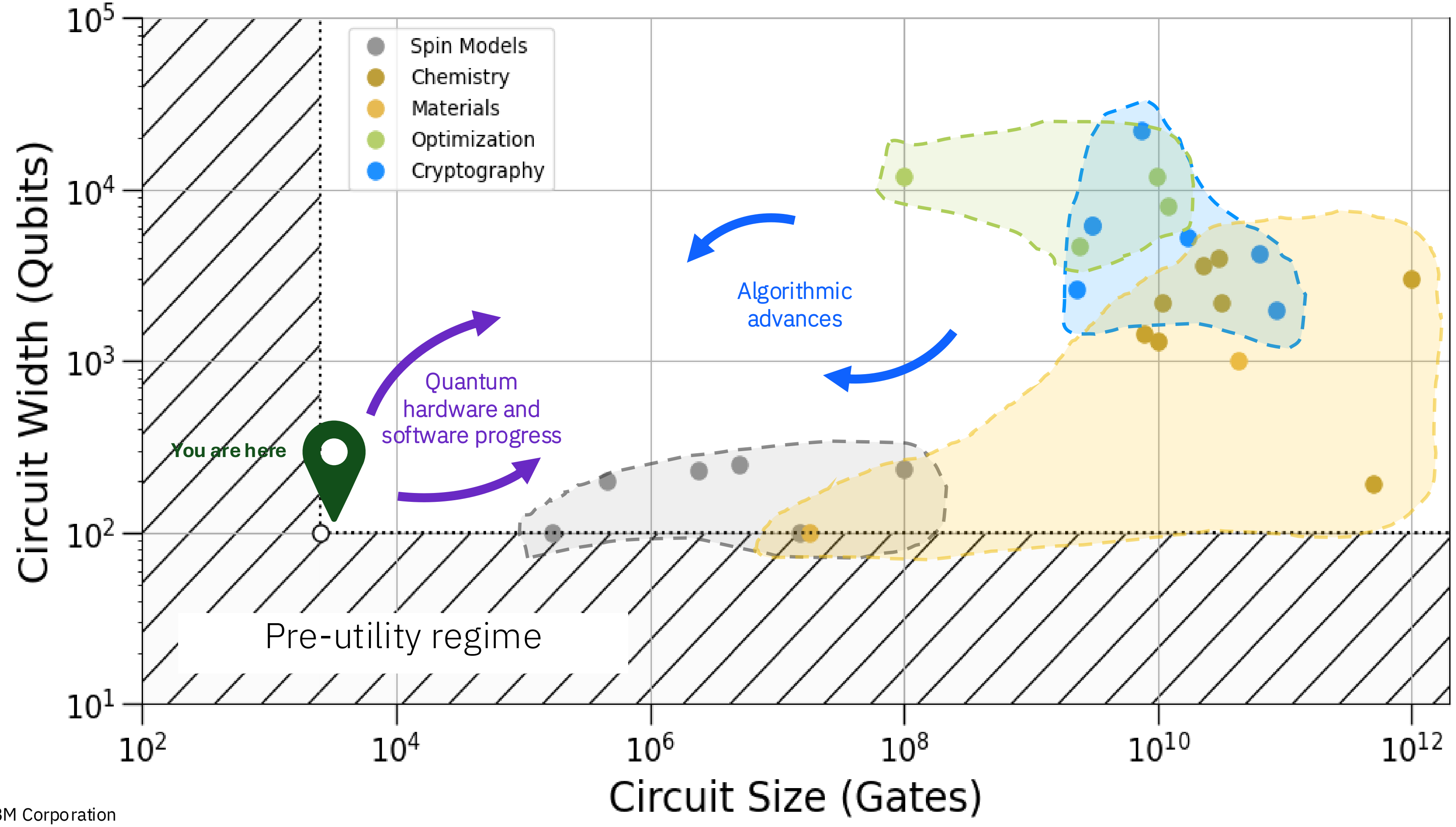
Useful Quantum = Hardware + Algorithm + Applications

Useful Work

QPUs

Qiskit

Quantum Software
(powered by Qiskit add-ons
and
Qiskit Functions)



The computationally complex problems exist across almost every industry

Banking
- Fraud monitoring
- Portfolio optimization
- Risk simulation
- Customer analytics
- Time series forecasting

Automotive
- Battery material design
- Material design
- Mobility as a Service
- Quality control
- Self-driving and ADAS
- Production optimization

Chemicals
- Sustainable products
- Low-carbon manufacturing
- Resilient supply chains
- Process optimization
- Asset health

Life sciences
- Efficient drug research and development
- Clinical trials
- Tractable protein folding
- Call-centric therapeutics
- mRNA

Healthcare
- Accelerated diagnoses
- Personalized interventions
- Adherence to drugs
- Biomarkers
- Image processing

Logistics
- Global logistics optimization
- Disruption management
- Routing optimization
- Predictive maintenance
- Forecasting

Public services
- Security/safety
- Multimodal transport
- City resource planning
- Disaster management
- Fraud detection in tax and social

Insurance
- Catastrophe modeling
- Precise customer profiling
- Efficient risk management
- Optimized pricing of premiums

Electronics
- Faster product design
- Circuit defect identification
- Process optimization
- Production optimization
- Quality control

Airlines
- Forecasting and revenue
- Irregular operations
- Network planning
- Safety and maintenance
- Hyper-personalization

Energy and utilities
- Energy trading
- Optimization of energy grid
- Renewables system design
- Energy forecasting
- Hyper-personalization
- Asset health

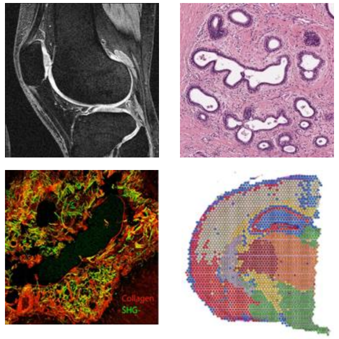
Aerospace
- Material discovery
- Aircraft design
- Asset health
- Corrosion and material interaction
- Fuel efficiency

Oil and gas
- Emissions reduction
- Reservoir simulation
- Virtual flow meters
- Subsurface modeling
- Failure prediction

Telecom
- Network optimization
- Network anomaly detection
- Contextual customer segmentation
- Cybersecurity network

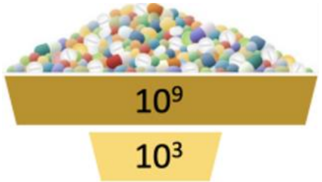
An Ecosystem of Quantum Algorithms for Healthcare and Life Sciences

Biomedical Imaging



- MRI denoising, super-resolution, segmentation, etc.
- X-ray/DXA scans
- Histopathology
- Second-Harmonic Generation

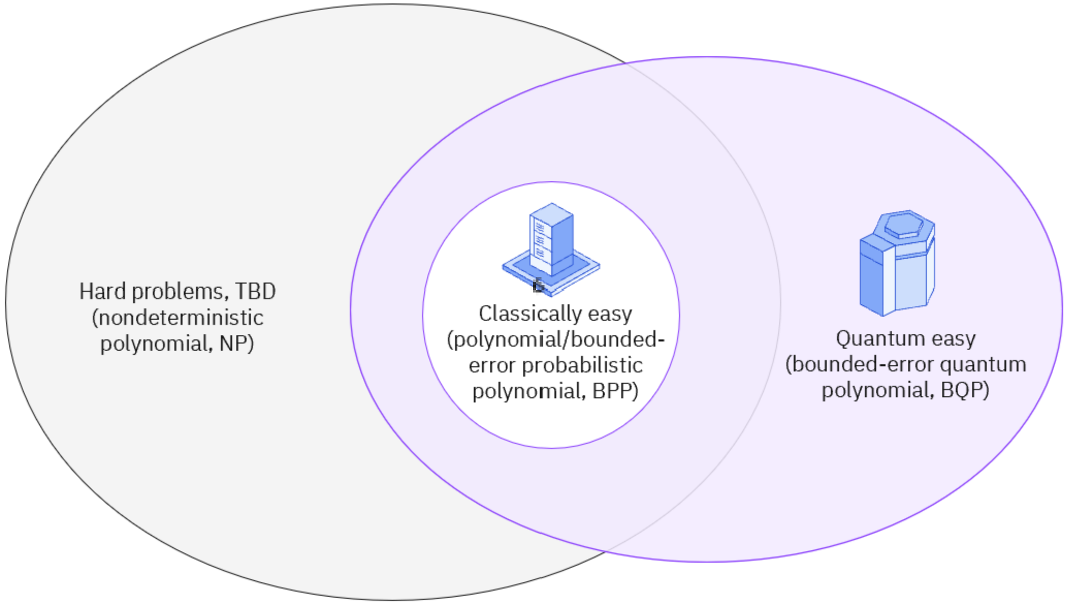
Virtual Screening



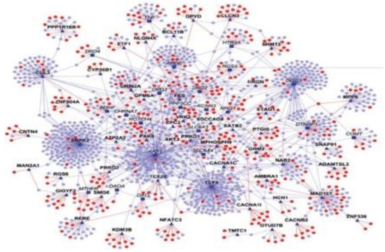
- Conformational Space Search
- Binding Affinity Prediction
- Post-Translational Modification
- Limited Experimental Data in ADMET

Cancer Immunotherapy

- Find potential signaling domains
- Accurately predict immunogenicity/cytoxicity.

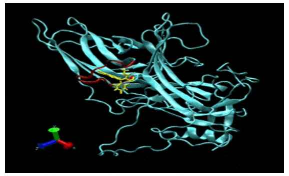


Network Biology



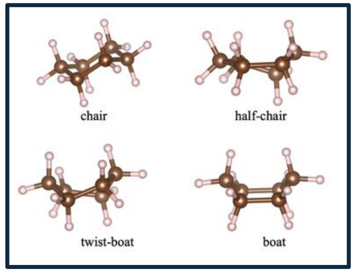
- Modeling Cellular Networks & Pathways
- Multiomics Analysis
- Genome Annotation
- Epigenetic Modification

Protein Design



- Structure Prediction
- De Novo Drug Design
- Modeling Allosteric Effects
- Polypharmacology

Chemistry



- Design specific inhibitors
- Find potential energy states
- Simulate molecules
- Drug Discovery

Pharmaceutical R&D Process

Discovery				Development			
Target ID & Verification	Hit Generation	Lead Gen & Optimization	Pre-Clinical	Phase 1 Safety	Phase 2 Safety & Efficacy	Phase 3 Safety & Efficacy	FDA Review & Approval
1-2 yrs	1-2 yrs	1-2 yrs	1 yr	1-2 yrs	1-2 yrs	2-3 yrs	1-2 yrs

IND submitted

NDA submitted

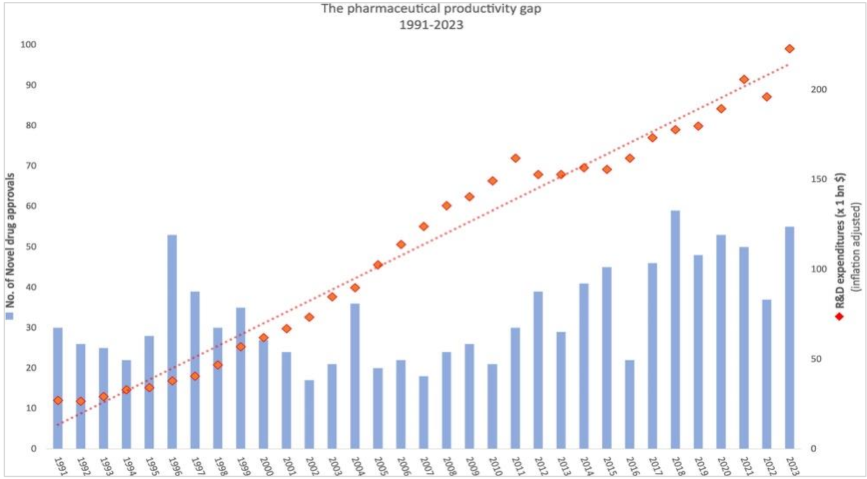
Drug Discovery & Development Challenges

$ 3B — Discovery Cost new drug 2024

10-15 Years — Time to Market

90% — Clinical drug development failure rate

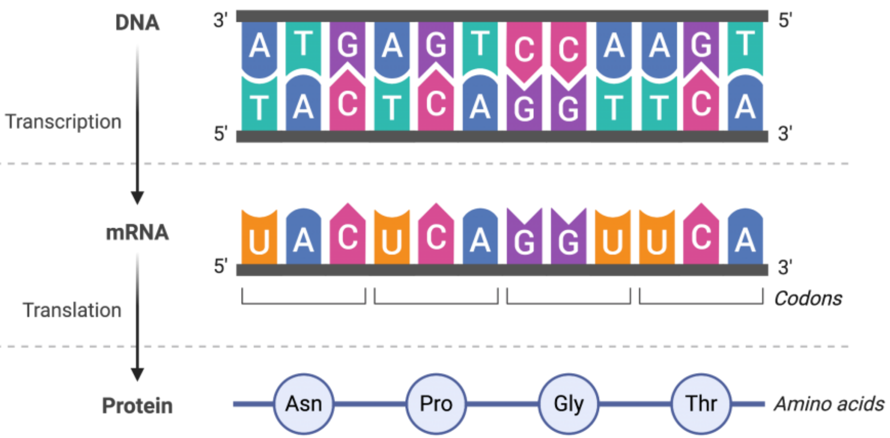
Source :https://doi.org/10.1016/j.apsb.2022.02.002



Source: https://doi.org/10.1016/j.drudis.2024.104160

Quantum Computing in Target Discovery

Target ID & Verification › Hit Generation › Lead Gen & Optimization › Pre-Clinical › Phase 1 Safety › Phase 2 Safety & Efficacy › Phase 3 Safety & Efficacy › FDA Review & Approval



Genomics

Transcriptomics

Proteomics

Metabolomics

Lipidomics

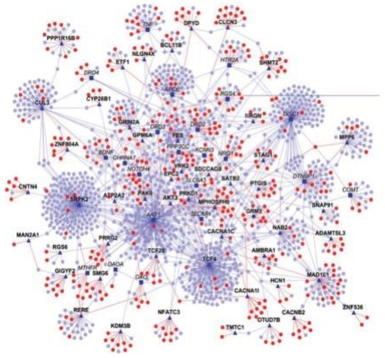
Epigenomics

Microbiomics

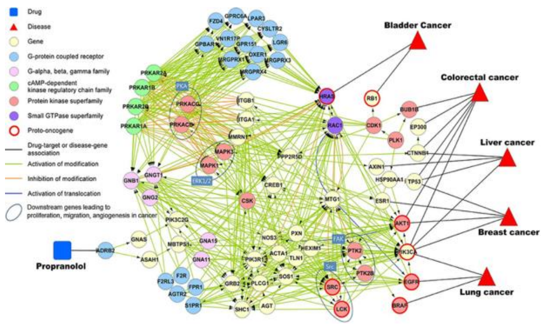
Pharmacogenomics

Cellomics

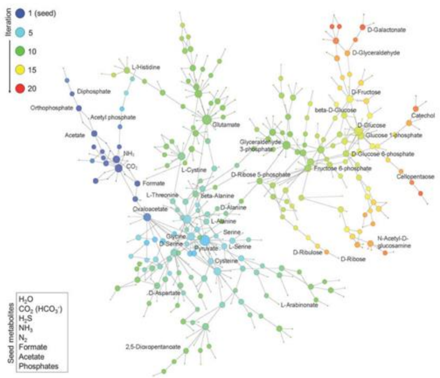
Many more …



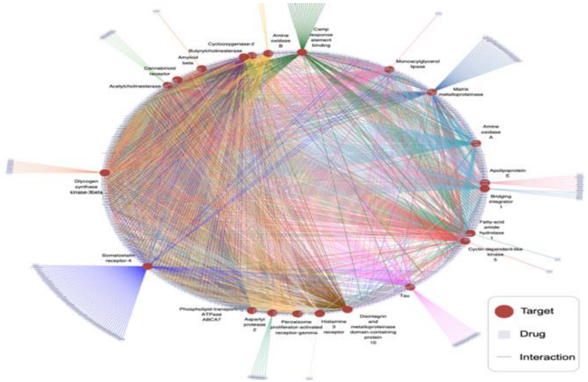
Human Reference Interactome (HuRI) map (Protein-Protein Interaction Network)



DNA–protein interaction networks

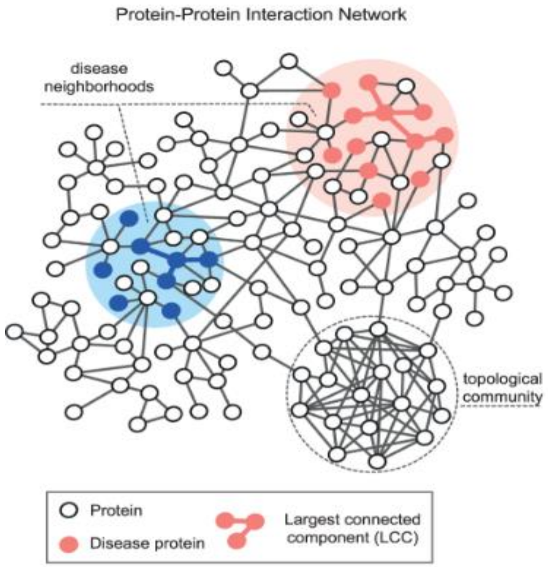


Metabolic networks



Drug-Target Network

Community Detection

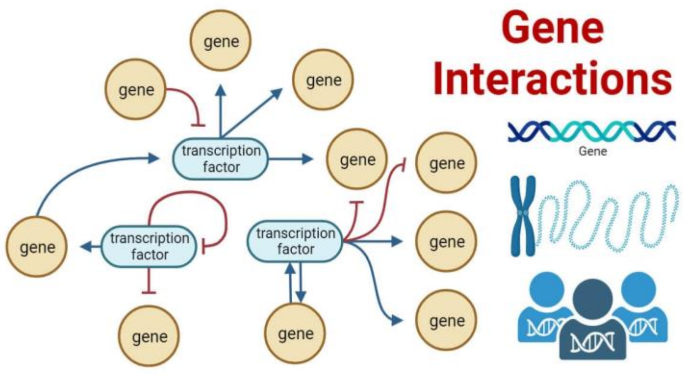


Problem: Classical methods struggle to identify critical proteins or genes in complex cancer signaling pathways like MAPK or PI3K, limiting discoveries of novel oncogenes or tumor suppressors.

Quantum Solution: Quantum walks can identify key proteins or genes in cancer signaling pathways, aiding in the discovery of new oncogenes or tumor suppressors for targeted therapies.

Ghiassian et al., https://doi.org/10.1371/journal.pcbi.1004120
Shaydulin et al., https://doi.org/10.1002/qute.201900029

Higher-order Epistasis

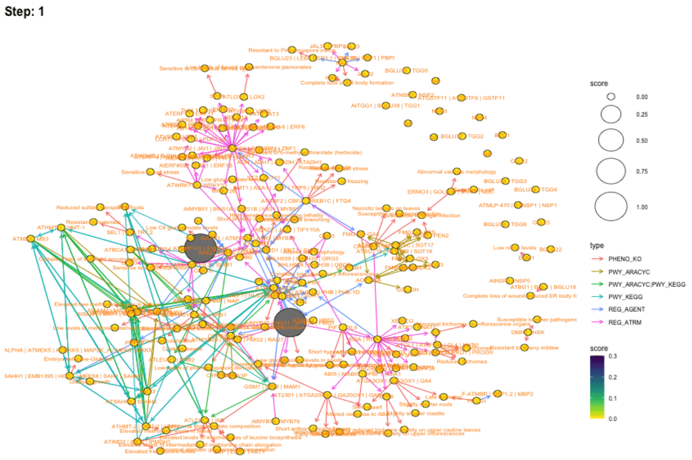


Problem: Polygenic diseases, like Type 2 diabetes and heart disease, involve multiple interacting genes, with higher-order epistasis (3+ gene interactions) revealing complex relationships that are challenging to analyze using classical methods.

Quantum Solution: By optimizing the discovery of gene interactions, quantum computing (Grover search based) can identify new disease pathways and therapeutic targets, accelerating drug development for polygenic diseases.

Source: https://microbenotes.com/gene-interactions/

Link Prediction

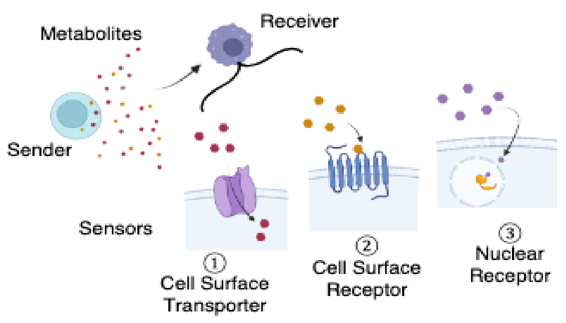


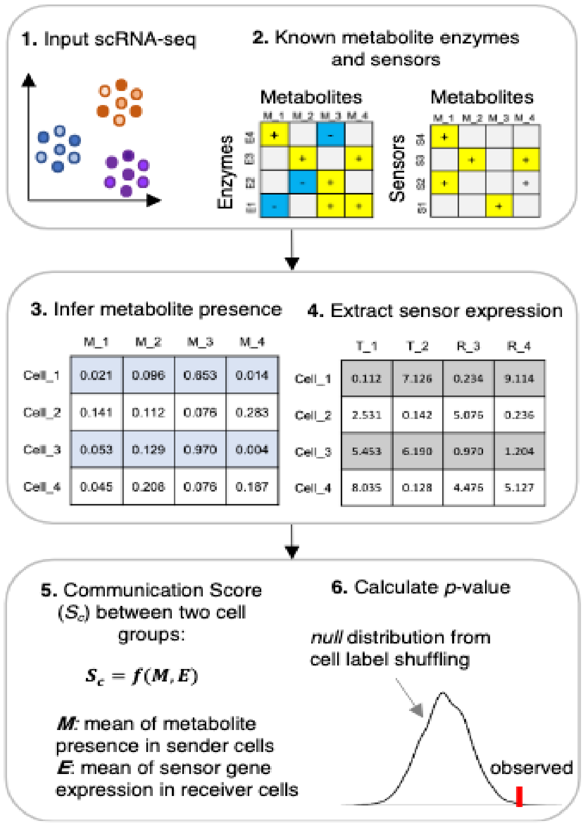
Problem: Link prediction in network biology involves identifying missing or potential interactions between biological entities, such as genes, proteins, or metabolic pathways, based on known data. Classical methods often struggle to handle the complexity and scale of biological networks

Quantum Solution: Quantum computers could enhance link prediction and biological target identification by leveraging quantum walks to explore networks more efficiently with greater speed and accuracy than classical systems, offering potential for improved drug-target discovery and network analysis

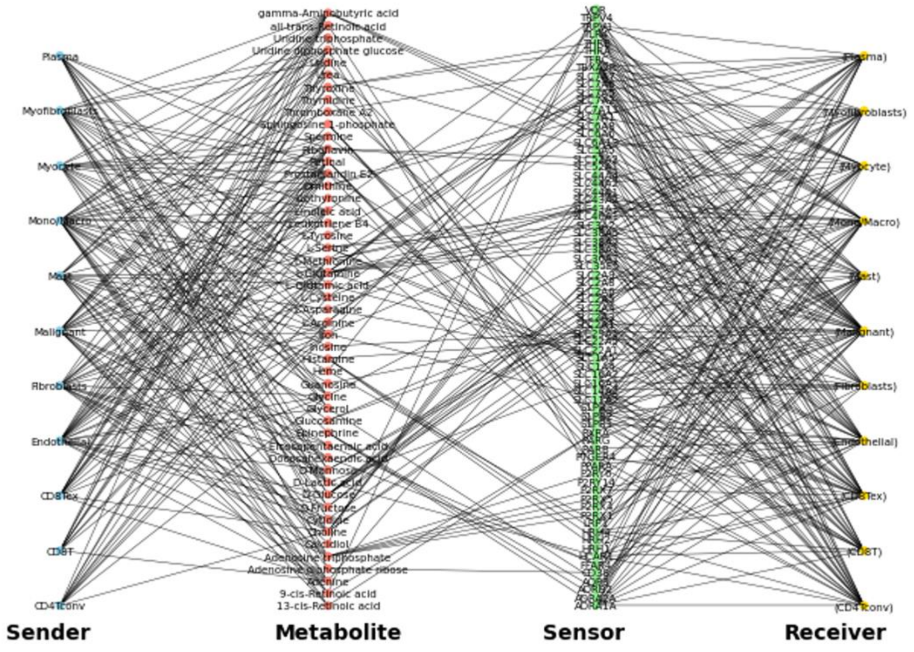
Source: David Kainer,
Saarinen et al., https://arxiv.org/abs/2311.05486
Dubovitskii et al. https://arxiv.org/abs/2506.06514

Quantum Random Wak in metabolic cell-cell interaction networks

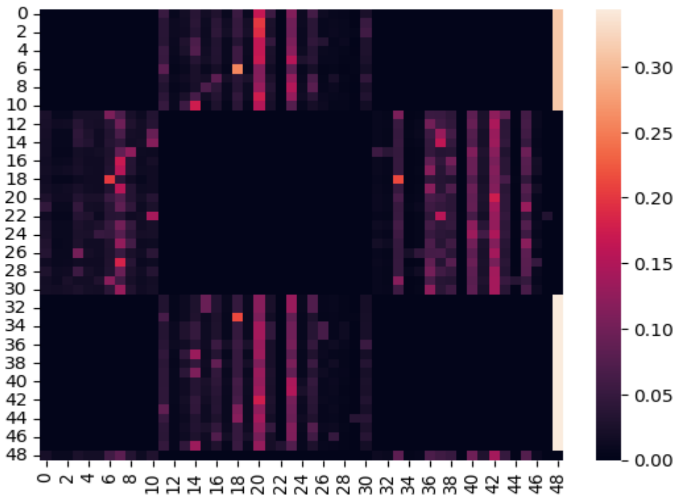




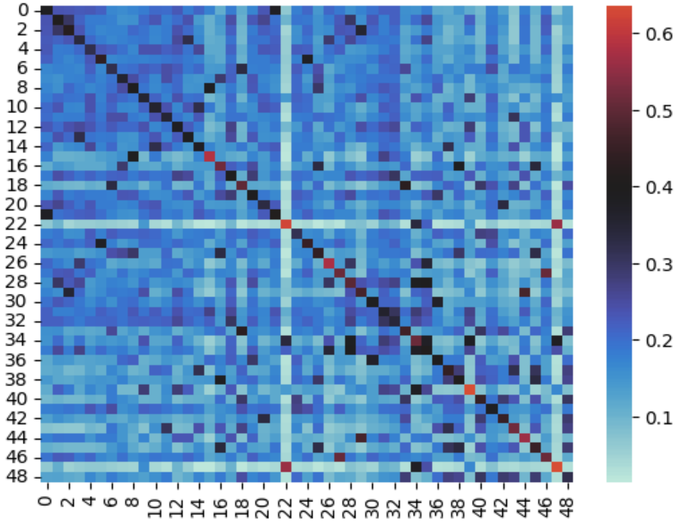
- Cell-cell interactions involving metabolites and sensor genes creates a multi-partite network from single-cell RNA-seq data of mouse brown adipose tissue (BAT)
- After five steps of evaluation the Euclidean distances between pairs of probability distributions recover five main communities in discrete-time random walk (DTRW), whereas in discrete-time quantum random walk (DT**Q**RW) there are more transitions between vertices in the CCI network.



Classical Random Walk



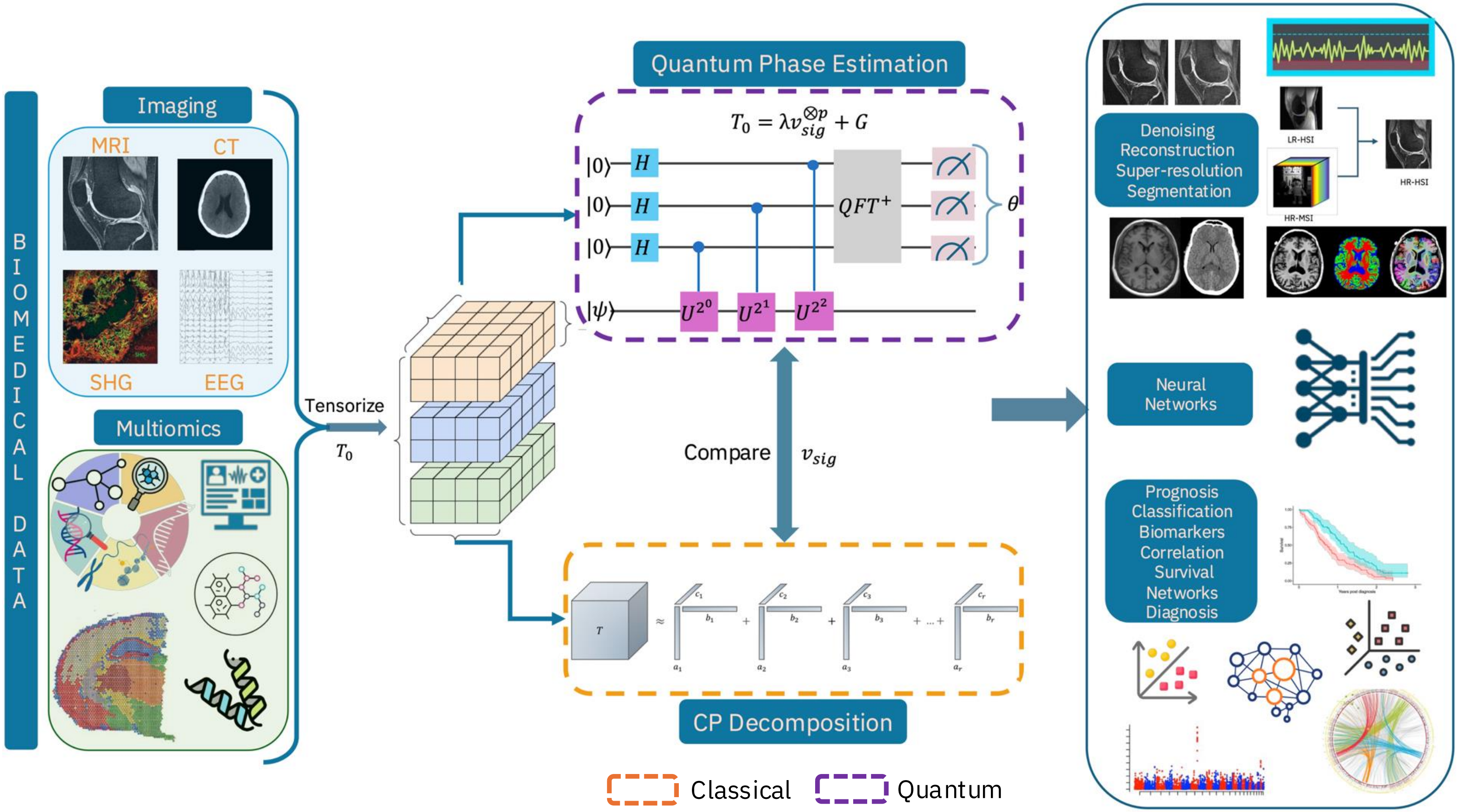
Quantum Random Walk

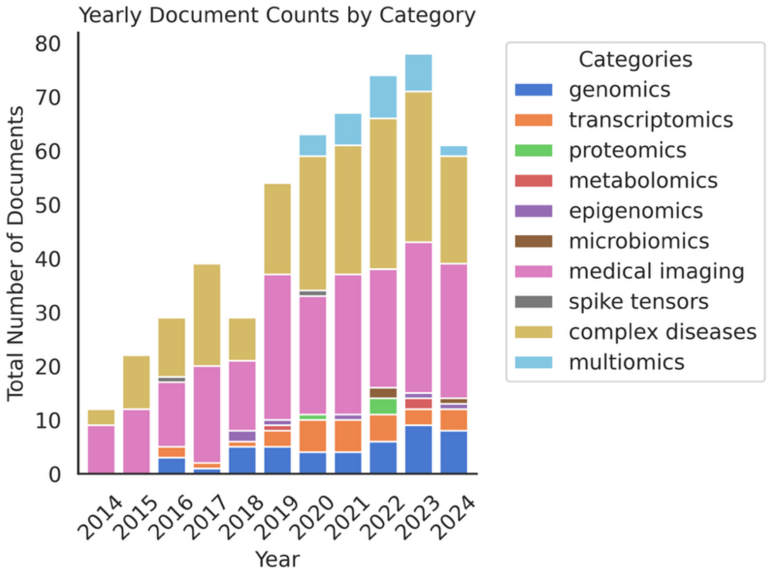


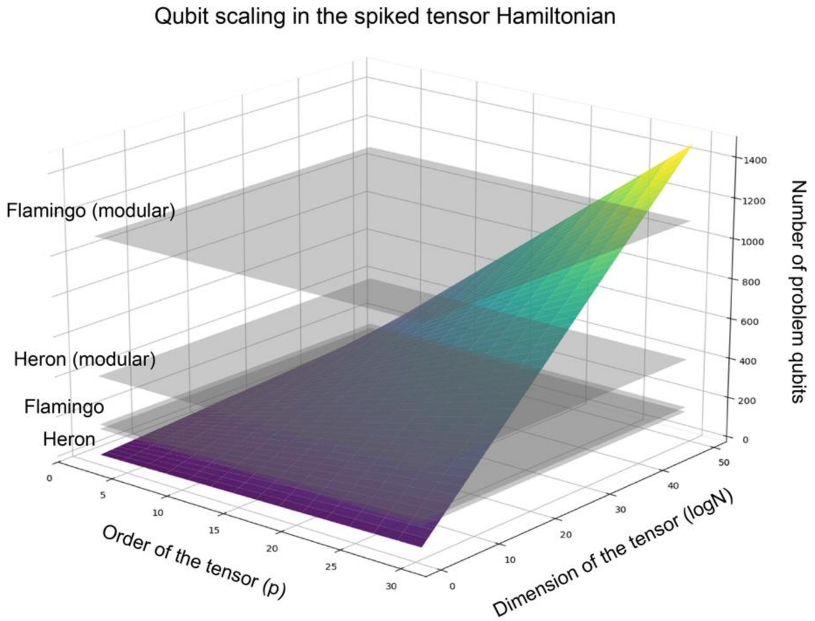
Dubovitskii et al. (2025) https://arxiv.org/abs/2506.06514

Quantum Tensor Decomposition for Biomedical Data

- Tensor decomposition is widely used in Biomedical data. It is NP-Hard and becomes intractable on a random tensor of size (N=100) and order > 4.
- Quantum tensor decomposition (QTD) has been shown to have a quartic speedup with provable guarantees. We are applying QTD on biomedical imaging data such as MRI, SHG, Spatial Transcriptomics, as well as in multi-omics data to accurately predict complex diseases.



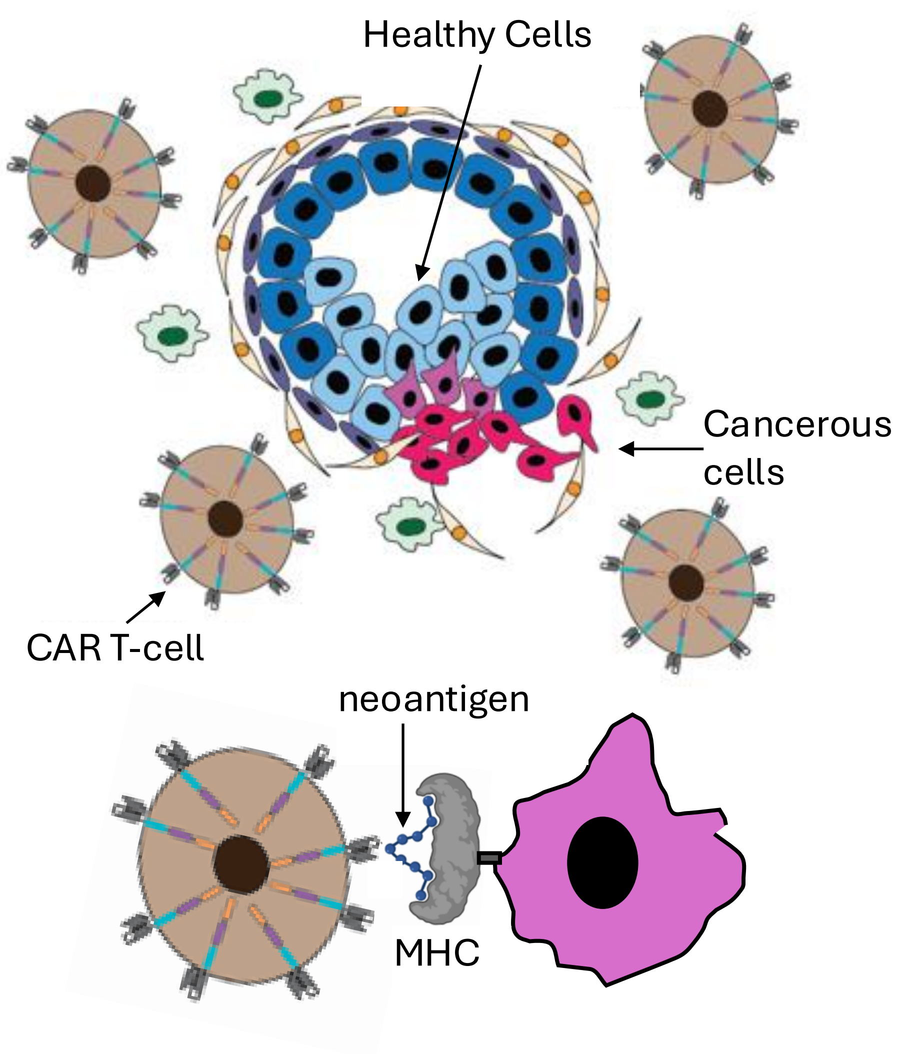




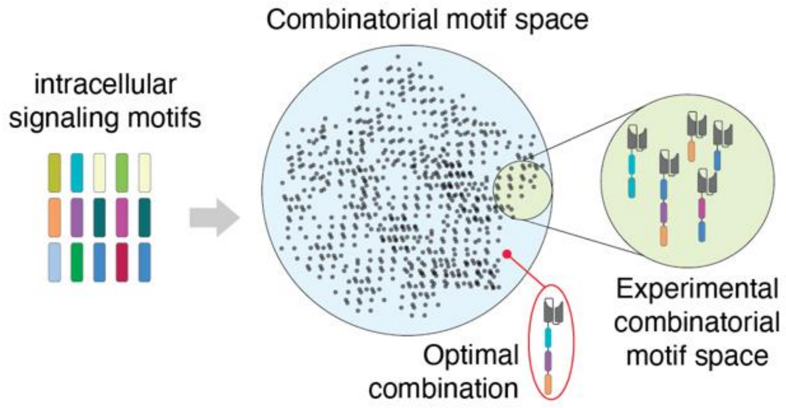
Burch et al. (2025) https://arxiv.org/abs/2502.13140

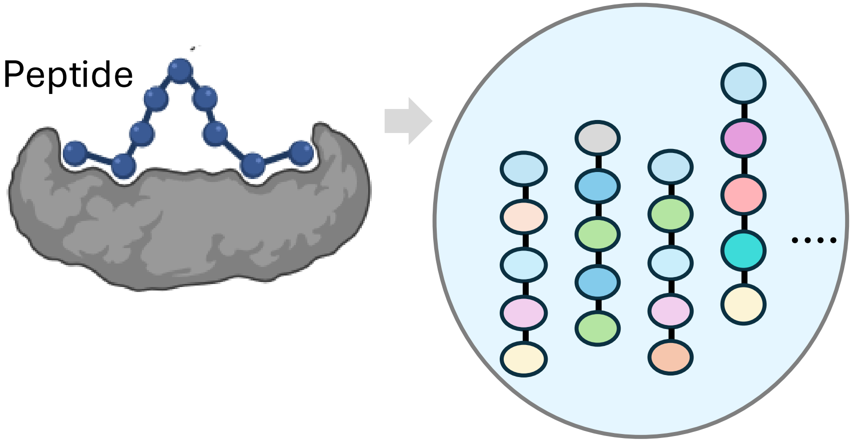
Quantum for ML – Applications in Immunotherapy

(A) Cell – protein interactions



(B) Data space grows combinatorially

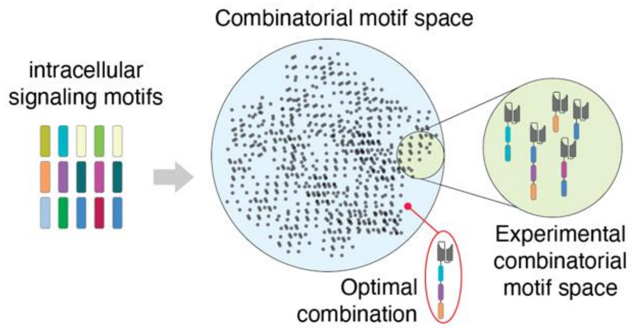




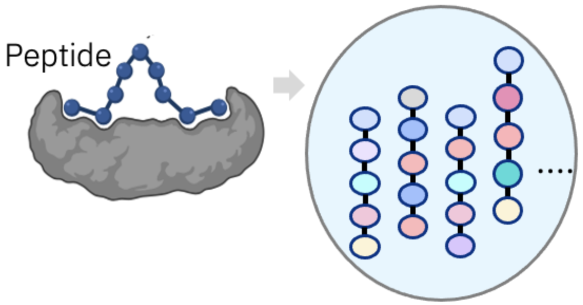
Data is experimentally generated; it is limited

Quantum for ML – Applications in Immunotherapy

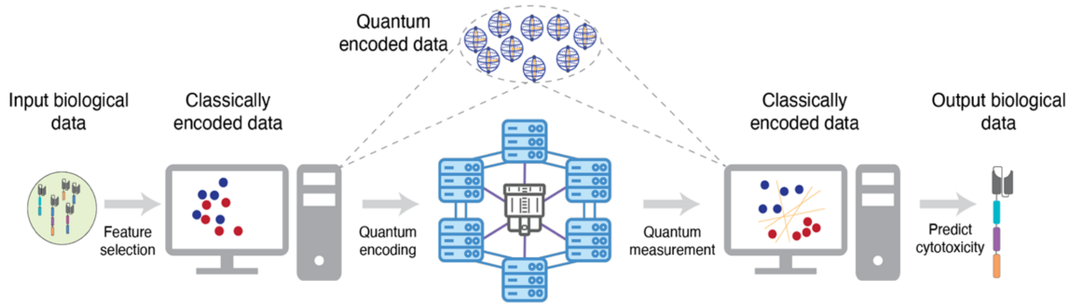
Use-case 1: Identification of cytotoxic motifs



Use-case 2: Predicting immunogenicity and binding

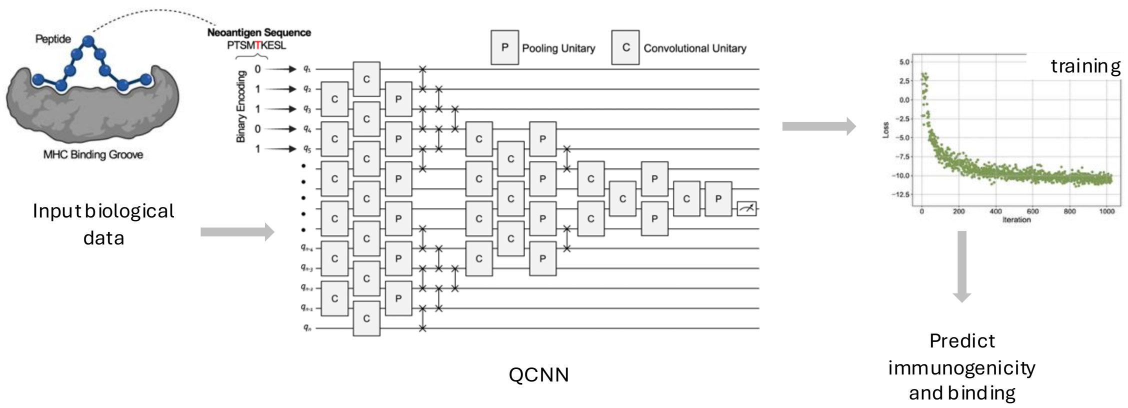


QML method: Projected Quantum Kernels



arXiv:2507.22710

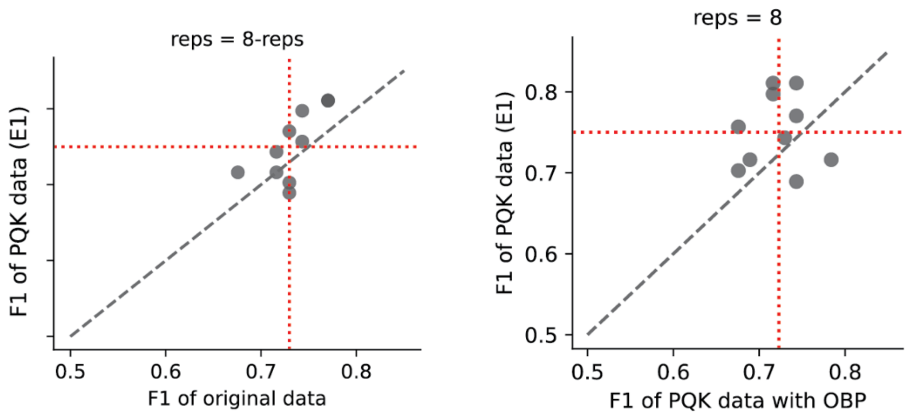
QML method: Quantum Convolutional Neural Networks



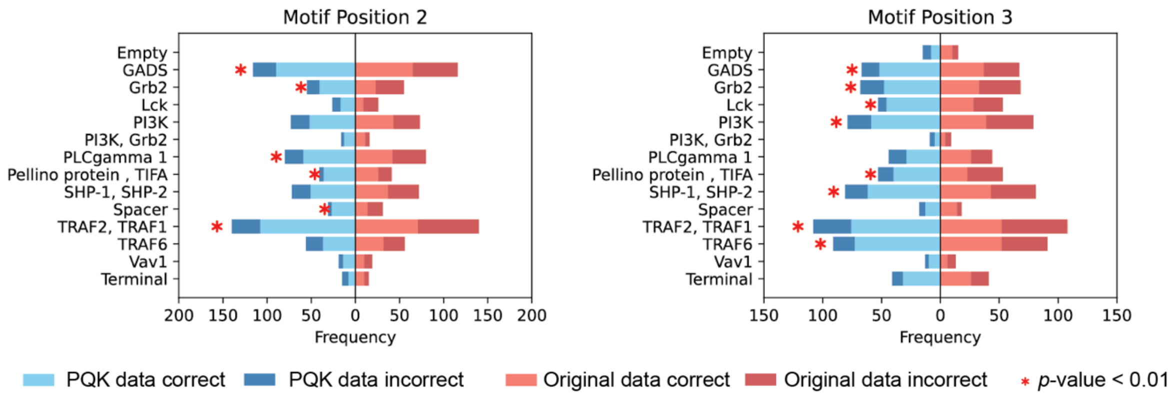
biorxiv:2025.07.29.667313v1

Quantum for ML – Applications in Immunotherapy

Use-case 1: Identification of cytotoxic motifs



Better prediction accuracy of the quantum model



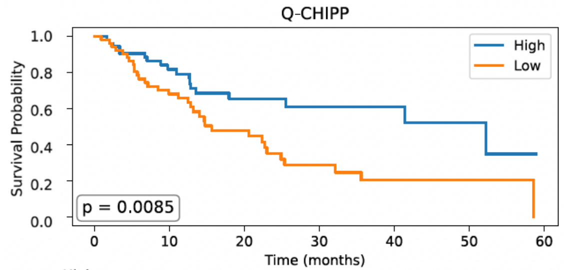
Quantum model provides unique biological insight

Utro et al. (2025) https://www.arxiv.org/abs/2507.22710

Use-case 2: Predicting immunogenicity and binding

Positions	QCNN	CNN (ltd.)	CNN (large)	RF
BA	**0.73**	**0.73**	**0.73**	0.68
1,4,5,9	**0.71**	0.33	0.67	0.64
1,4,5,9,BA	0.71	0.50	**0.72**	**0.72**
4,6,7,8	0.49	0.39	0.33	**0.52**
4,6,7,8,BA	0.54	0.35	0.68	**0.72**
4,6,8,9	0.70	0.35	**0.71**	0.68
4,6,8,9,BA	0.74	0.55	0.71	**0.77**
Full peptide	**0.70** *	0.60	0.65	0.66

QCNN model is better or comparable to classical ML models



	0	10	20	30	40	50	60
High							
At risk	53	32	19	11	7	3	0
Censored	0	12	19	26	30	33	35
Events	0	9	15	16	16	17	18
Low							
At risk	52	32	16	8	4	2	0
Censored	0	4	12	14	16	18	19
Events	0	16	24	30	32	32	33

Survival predicted by the quantum model beats classical predictions

Peters et al. (2025) https://doi.org/10.1101/2025.07.29.667313

Quantum Computing in Virtual Screening of Drug Candidates

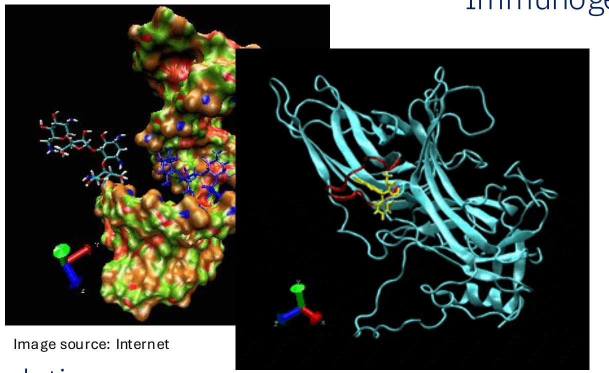
Target ID & Verification > Hit Generation > Lead Gen & Optimization > Pre-Clinical > Phase 1 Safety > Phase 2 Safety & Efficacy > Phase 3 Safety & Efficacy > FDA Review & Approval

Clinical Development

Stability

Immunogenicity

Drug Delivery



Binding Affinity

Image source: Internet

Formulation Development

Synthesizability

- Small molecules
- Oligonucleotides
- Peptides
- Enzymes
- Antibodies
- Cell Therapy
- Stem Cell

Biologics

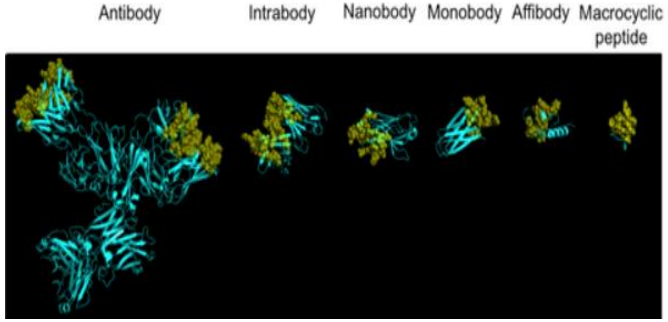


Image source: Internet

Chemical Space: 20^{K}

K = number of residues

Classical computer

With a Force Field:

Less Accuracy

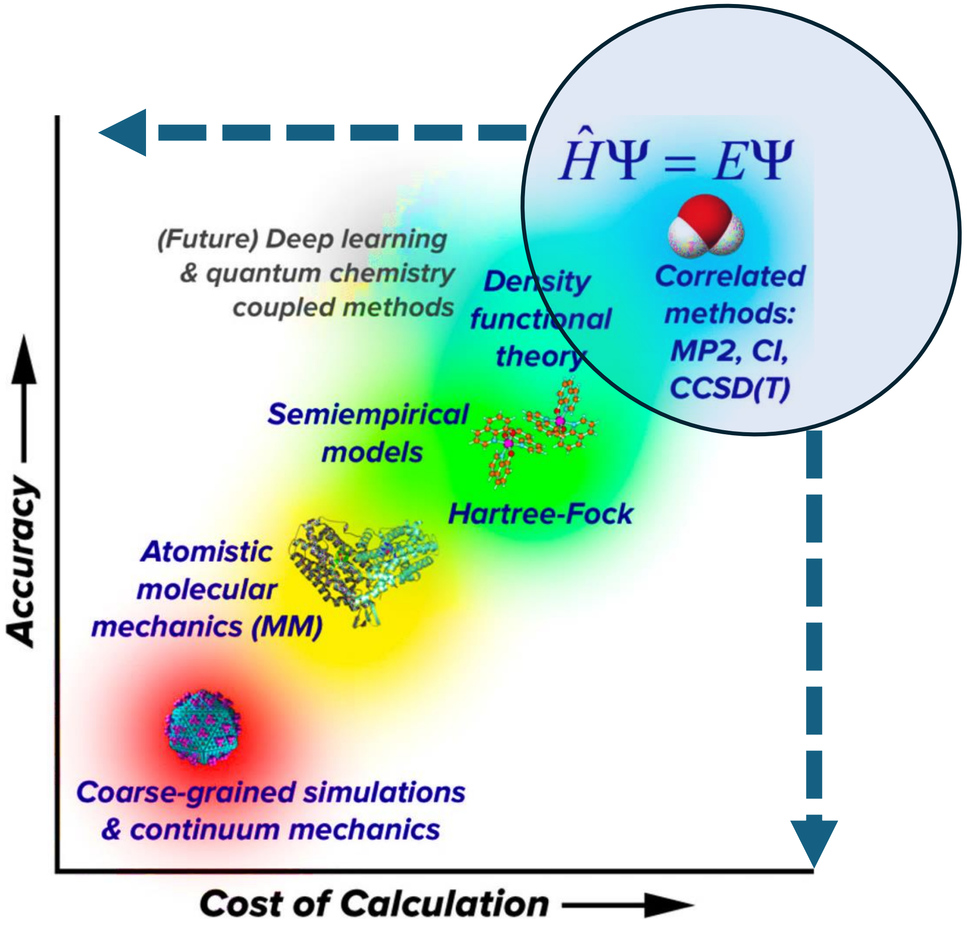
Quantum computer

With Quantum Simulations:

More Accuracy

Multi constraint – multi-objective optimisation problem

Quantum Computing in Virtual Screening of Drug Candidates



Source: Chem. Rev. 2021, 121, 10, 5633-5670

FT

- Quantum Phase Estimation
- Quantum Amplitude Estimation

NISQ

- Variational Quantum Eigensolver
- Quantum Approximate Optimization Algorithm

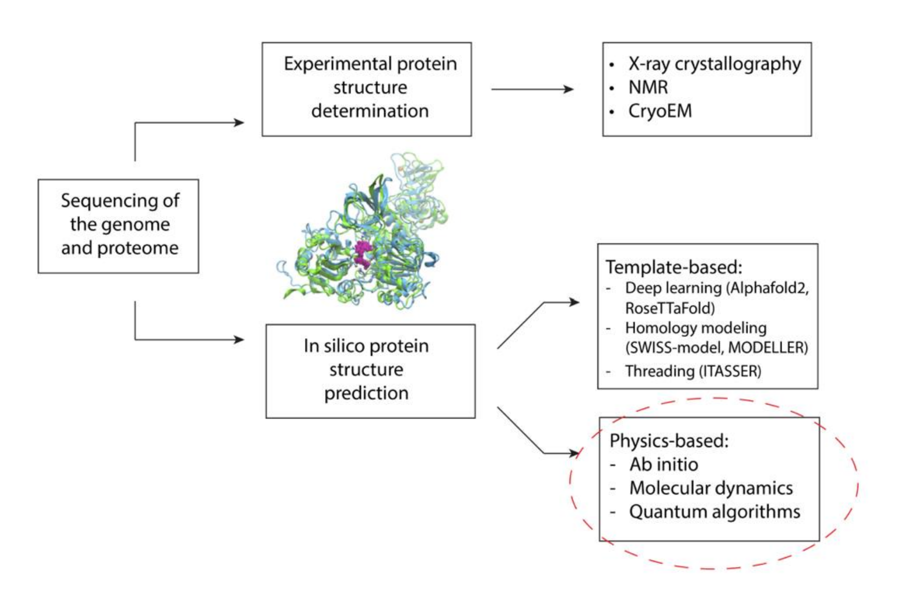
Quantum Computing Applications

- Accurate Simulation of Supramolecular Interactions
- Accurate Binding Energy Calculation
- Ligand Design & Lead Optimization

Case Study with Cleveland Clinic (Protein Folding)

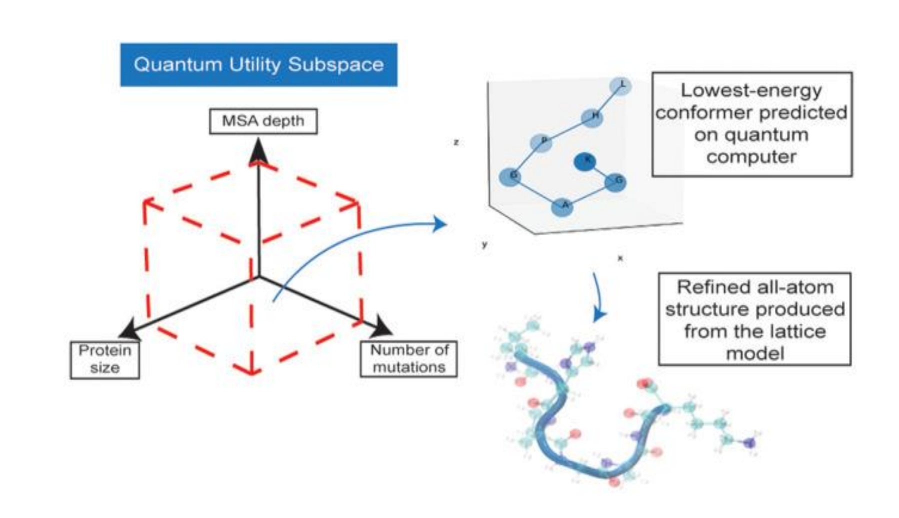
Problem

Predicting protein structures is critical for creating new biological products, but classical deep learning struggles with mutated sequences, IDRs, and low MSA depth.



Quantum Method

Hybrid quantum-classical algorithms, like VQE and QAOA, improve native protein structure identification, addressing classical limitations.



Doga et al. https://arxiv.org/abs/2312.00875
Robert et al., https://www.nature.com/articles/s41534-021-00368-4

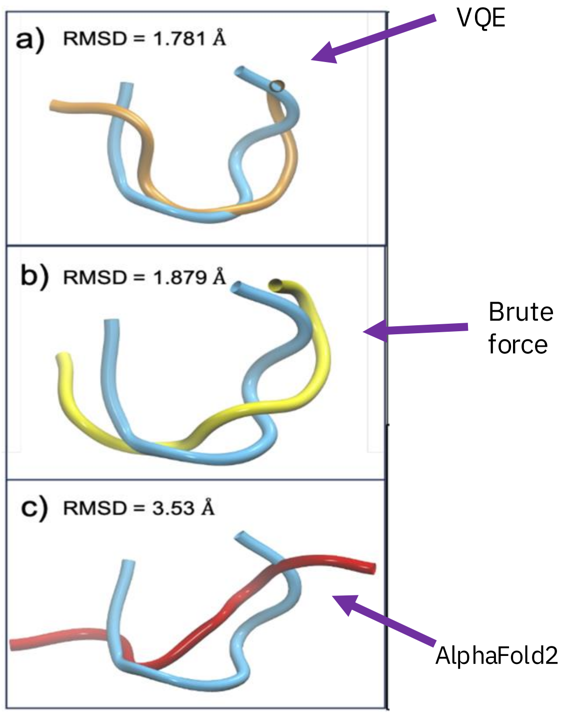
Case Study with Cleveland Clinic (Protein Folding)

Method:

VQE model is used to accurately predict the structure of a 7 amino acid catalytic loop of the Zika Virus NS3 Helicase.

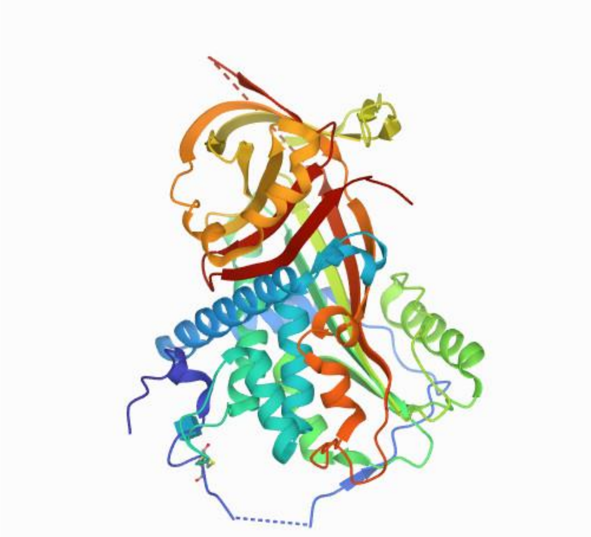
Solution:

Quantum optimization can advance protein structure prediction, driving innovations in biological product design and therapeutic development.



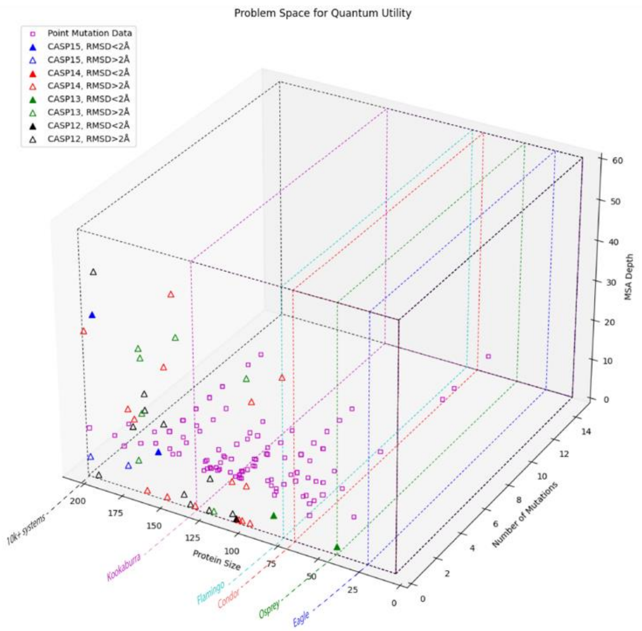
7 amino acid protein is mapped to 9 qubits on the IBM Cleveland quantum computer

Doga et al. https://pubs.acs.org/doi/10.1021/acs.jctc.4c00067



10 amino acid Angiotensin peptide on 22 qubits

Robert, A. et al. https://doi.org/10.1038/s41534-021-00368-4



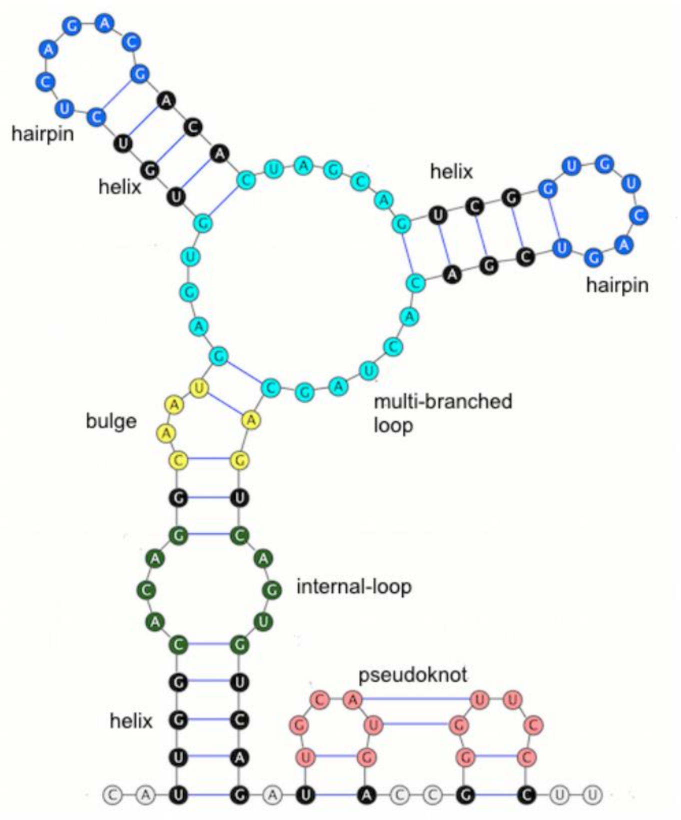
Case Study with Moderna (mRNA Structures Prediction)

Problem:

Predicting mRNA secondary structures is a formidable combinatorial optimization problem due to the vast number of possible configurations and their associated energy states.

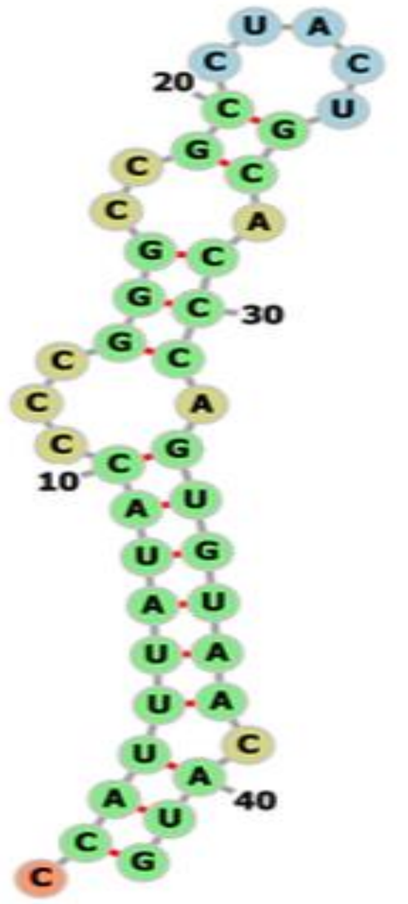
Solution:

Results indicate that quantum computing offers valuable insights into RNA folding mechanisms and enables rapid identification of functional RNA structures. This work lays the groundwork for future research in quantum-assisted bioinformatics and highlights the importance of developing scalable quantum algorithms for real-world biological problems.



An RNA secondary structure illustrating five kinds of secondary structural elements: hairpin loops (blue), bulge (yellow), helix (black), internal loop (green) and multi-branched loop (cyan) and a pseudoknot (pink)

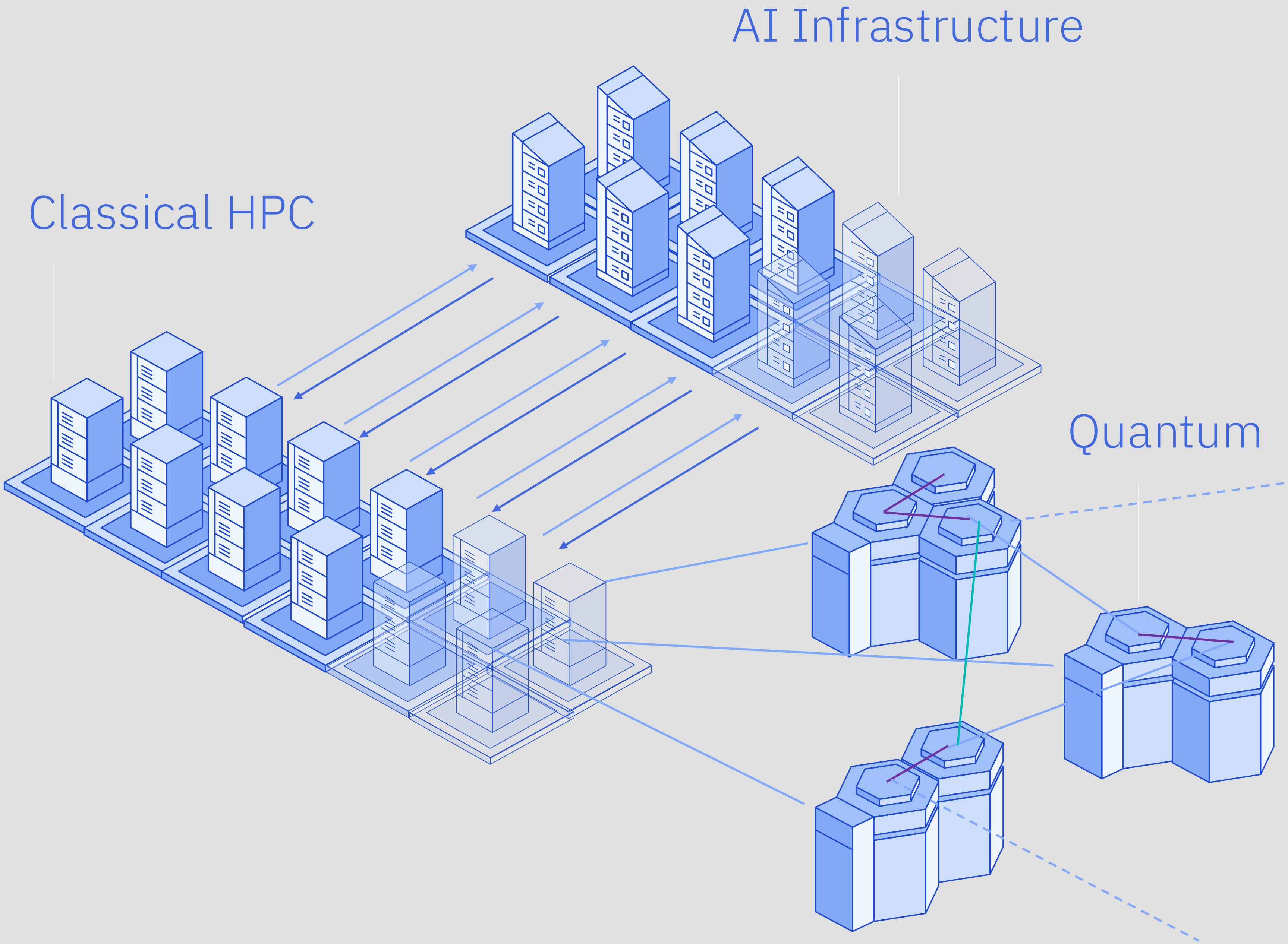
Alevras et al., https://arxiv.org/pdf/2405.20328



Optimal folded structure of the 42- nucleotide mRNA sequence on 80-qubit based on the corresponding lowest energy bitstring found by the hardware run.

Next Generation Computing

Bits + Neurons + Qubits

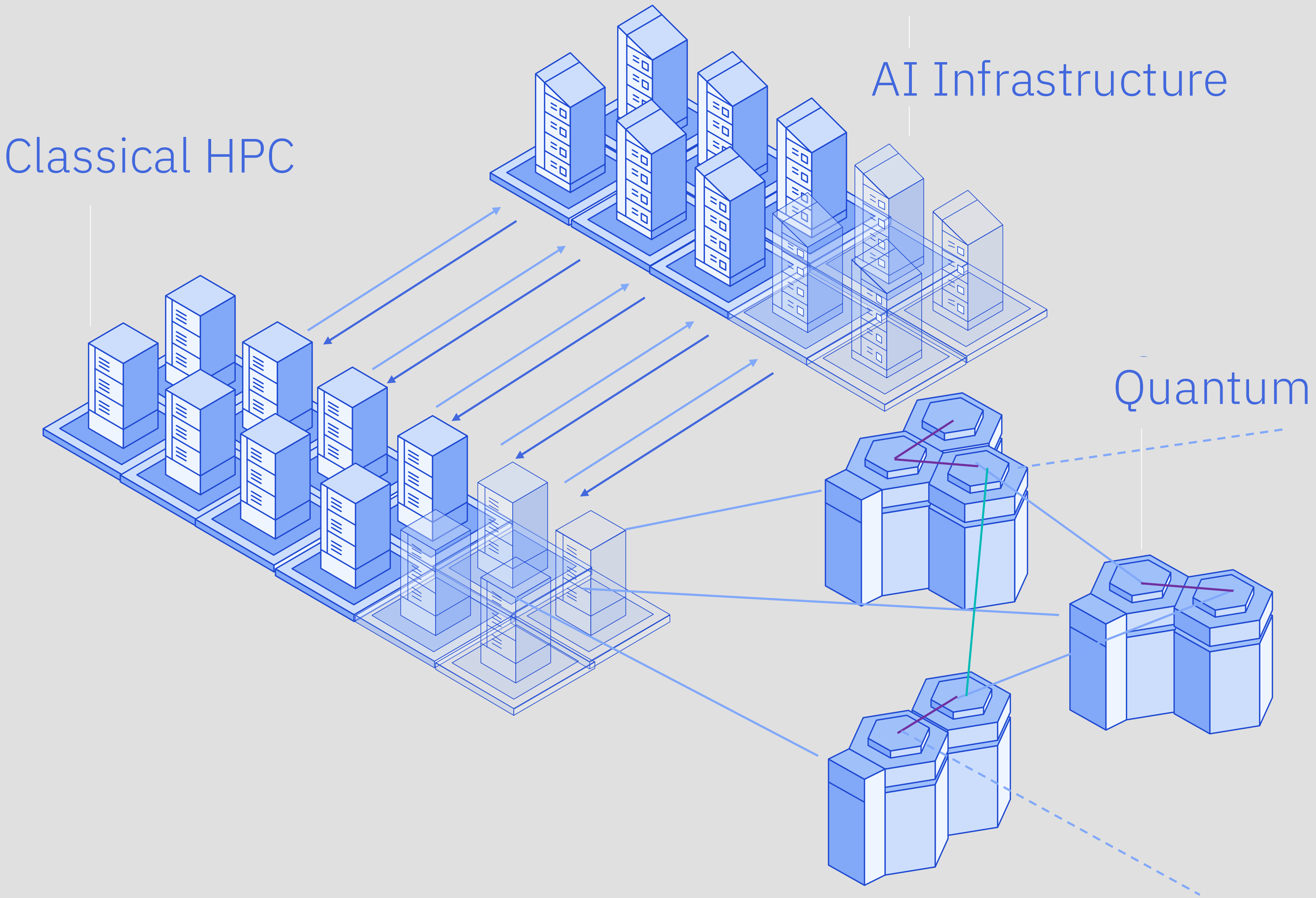


Bits + Neurons + Qubits

IBM's Quantum-Centric Supercomputing (QCSC) — merges classical and quantum systems, paving the way for solving pharma-relevant problems that are intractable today — with chemistry quantum advantage expected by 2026.

Early quantum integration builds strategic advantage — enabling pharma companies to shorten the learning curve, develop in-house quantum expertise, and position themselves as leaders in next-gen R&D innovation.

Future of Computing: Quantum Centric Supercomputing



Quantum-centric Supercomputing Integration at RIKEN (R-CCS)

RIKEN Fugaku supercomputer

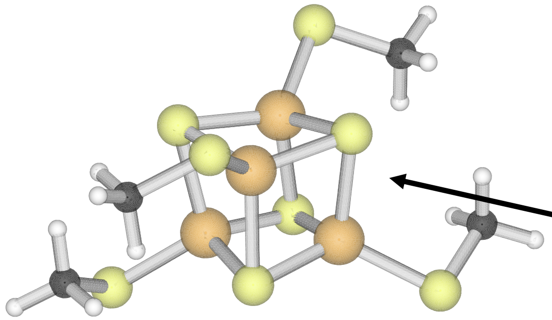
IBM Quantum System Two

Planned deployment 2Q 2025
Infiniband 400 Gb/s
Co-located

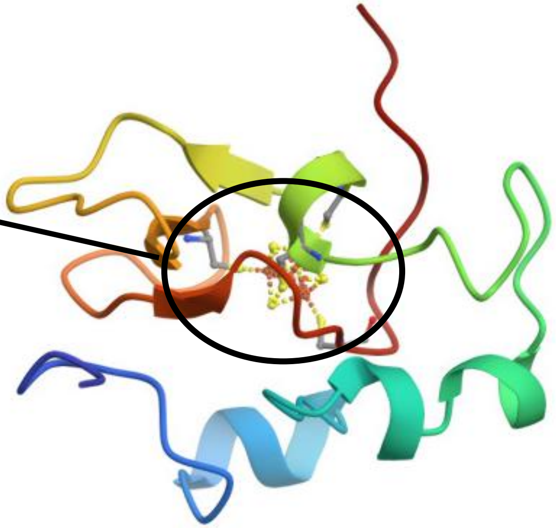
⟷

Quantum Chemistry on Pre-Fault-Tolerant Devices:

Sample-based quantum diagonalization (SQD)



[4Fe-4S] on a 36-orbital active space:
72 qubits



Fault-tolerant	Phase estimation qubits: **4.53M** **13 days** *
Pre-fault tolerant	VQE estimation at 10μs/circuit ~**3M years**
Quantum-centric supercomputing	Subspace estimation at 10μs/circuit ~**2 hours**

Applications in Drug Design

Understanding Protein Function
Designing Specific Inhibitors
Predicting Cluster Reactivity

*Estimation done using surface code

Density Matrix Embedding Theory (DMET) + SQD

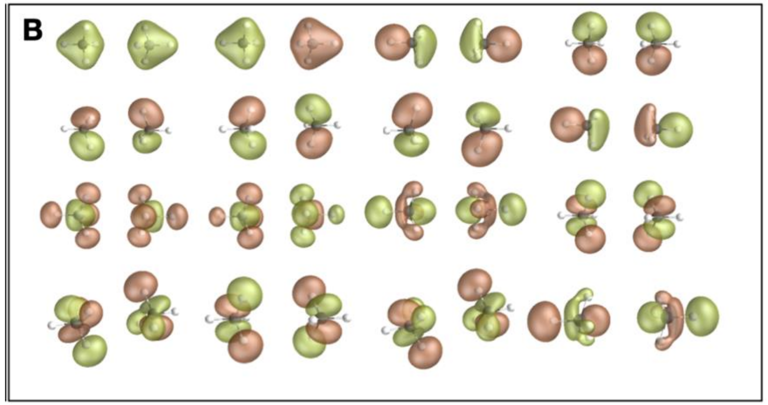
arXiv:2405.05068 (2024)

https://www.ibm.com/quantum/case-studies/modeling-realistic-chemistry#demo

Industry Applications of Sample-based Quantum Diagonalization (SQD)

Cleveland Clinic

Simulate the potential energy surfaces (PES) of the water and methane dimers (hydrophilic and hydrophobic interactions)

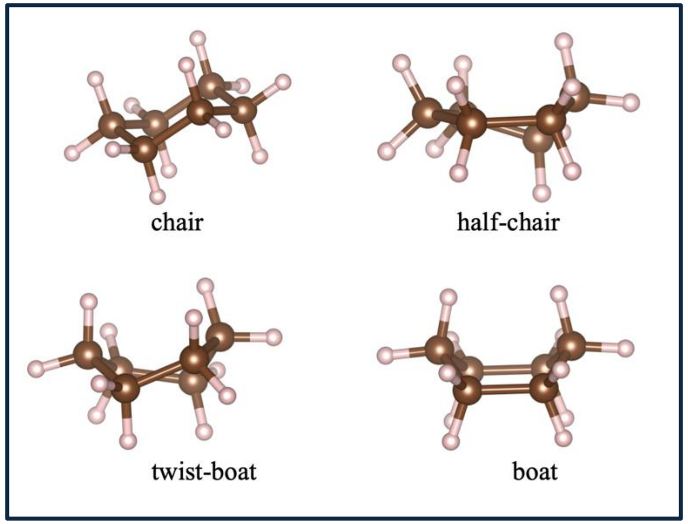


54 qubits, 1600 2Q gates

arXiv:2410.09209

Cleveland Clinic

Simulation of relative energies of cyclohexene conformations



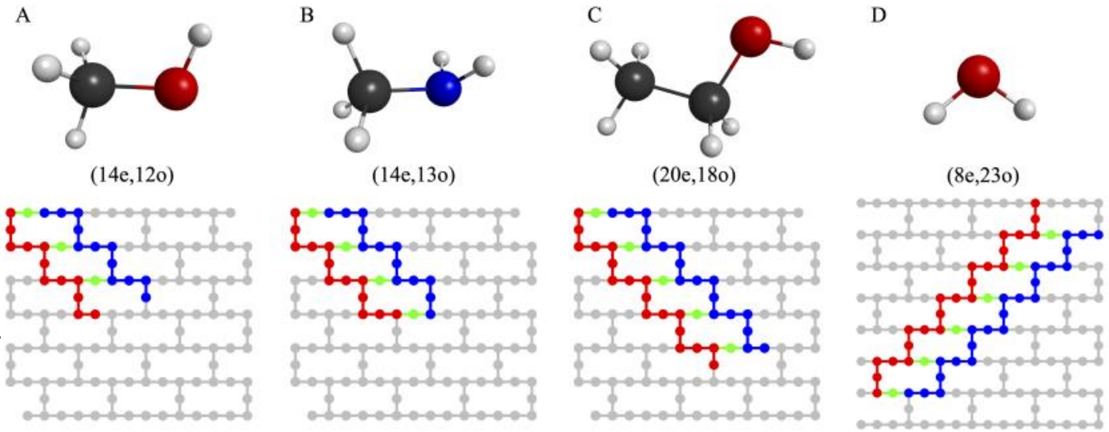
41 qubits, 1500 2Q gates

arXiv:2411.09861

Cleveland Clinic

SQD based simulation of
A. Methanol
B. Methylamine
C. Ethanol
D. Water

Qubit layouts of Local Unitary Cluster Jastrow (LUCJ) circuits. Qubits used to encode occupation numbers of spin-up/down electrons are marked in red/blue, while ancilla qubits denoted in green.



27 qubits | 30 qubits | 41 qubits | 52 qubits

arXiv:2502.10189

Thank you.